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Evaluation of CT-Guided Pudendal Nerve Cryoneurolysis for the Treatment of Refractory Pelvic Pain

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Abstract

Background: Pelvic pain caused by malignant and non-malignant etiologies can be treated by cryoneurolysis of the pudendal nerve(s). The purpose of this report is to assess the efficacy and safety of CT-guided pudendal nerve cryoneurolysis. Additionally, treatment outcomes were compared between malignant and non-malignant pelvic pain etiologies.

Methods: Patients who underwent CT-guided pudendal nerve cryoneurolysis for intractable pelvic pain were retrospectively reviewed. Pre-procedural and post-procedural pain scores, supplemental analgesic use, and complications were assessed.

Results: A total of 30 patients underwent CT-guided pudendal nerve cryoneurolysis. Pre-treatment Visual Analog Scale (VAS) pain scores were significantly improved at 1 week (3.9 ± 2.1, $P < 0.001$), 1 month (4.6 ± 2.3, $P < 0.001$), 3 months (3.9 ± 2.2, $P < 0.001$) and 6 months (6.2 ± 3.4, $P < 0.001$) post-treatment. Pre-treatment morphine equivalents (ME) significantly decreased at 1 week (30 (range:0-216), $P = 0.04$) and 1 month (17.5 (range:0-216), $P = 0.032$) post-treatment. Both malignant and non-malignant cohorts demonstrated significant improvement of VAS score at all time points; however, there was no significant difference in numeric improvement or percentage improvement in VAS score between the 2 cohorts. No adverse events were observed.

Conclusion: CT-guided pudendal nerve cryoneurolysis is an effective and safe treatment modality in patients with refractory pelvic pain from both malignant and non-malignant etiologies.

Keywords

cryoneurolysis, cryoablation, pudendal nerve, pelvic pain, interventional pain, nerve ablation

Introduction

Chronic pelvic pain may affect up to 15% of the U.S. population and is associated with an estimated annual health-care cost of approximately $5.8 billion.[1,2] Intractable pelvic pain despite conservative treatment is a challenging clinical condition that may arise from both malignant and nonmalignant etiologies. Malignant pain is often severe and progressive, with up to 30% of patients not achieving adequate relief with conventional systemic therapy.[3] Nonmalignant chronic pelvic pain, which may arise from multiple pelvic pathologies, is associated with a marked reduction in quality of life.[1,2,4]

Considering the high failure rate of conservative pain management and the opioid epidemic, alternative pain interventions focused on the neural system and pain transmission have gained attention.[5-8] Cryoneurolysis, in particular, has received increased interest because it provides pain relief through reversible nerve injury.[6,8] Pudendal nerve cryoneurolysis may be an effective option for both malignant and non-malignant pelvic pain, as the pudendal nerve innervates the anus, perineum, vagina, and deeper pelvic structures.[7,9] Computed tomography (CT)-guided pudendal cryoneurolysis may therefore offer a safe and valuable option for patients with intractable pelvic pain.

To date, CT-guided pudendal nerve cryoneurolysis has been evaluated only in a limited number of studies, predominantly case reports and small series[7-9]. The purpose of the

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Table 1. Demographic Data

Variable	Pudendal (n = 30)
Age (years)	59.6 ± 12.6
Admission status	
Inpatient	12 (12/30, 40%)
Outpatient	18 (18/30, 60%)
Sex	
Male	11 (11/30, 36.7%)
Female	19 (19/30, 63.3%)
INR	1.2 ± 0.2
Hemoglobin (g/dL)	10.6 ± 3.2
Creatinine (mg/dL)	0.9 ± 0.7
Platelets (K/uL)	291.7 ± 137.4
Bilirubin (mg/dL)*	0.3 (range:0.2-3)
Cause of pain	
Malignant	25 (25/30, 83.3%)
Non-malignant	5 (5/30, 16.7%%)
ECOG*	2 (range:0-3)
Pretreatment block	
Yes	20 (20/30, 66.7%)
No	10 (10/30, 33.3%))

INR = international normalization ratio, ECOG = Eastern Cooperative Oncology Group performance status.

present study is to assess the safety and efficacy of percutaneous CT-guided pudendal nerve cryoneurolysis in patients with intractable pelvic pain and to compare treatment outcomes between malignant and non-malignant pelvic pain etiologies.

Materials and Methods

This is a single-center retrospective review of consecutive patients who underwent CT-guided pudendal nerve cryoneurolysis between March 2022 and September 2025. The study was approved by the local Institutional Review Board and complied with the Health Insurance Portability and Accountability Act. Patients who did not have a Visual Analog Scale (VAS) pain score prior to and at 1 week and/or 1 month after the procedure (n = 0) and those who underwent block but not cryoneurolysis (n = 2) were excluded.

A total of 30 patients were identified who had undergone CT-guided pudendal cryoneurolysis. Preprocedural and postprocedural VAS pain scores, morphine equivalents (MEs), and adverse events (AEs) were collected from the available inpatient and outpatient medical records at 1 week, 1 month, 3 months and 6 months. VAS pain scores were taken from notes during clinic follow up with performing Interventional Radiologist or Palliative Medicine physicians and visits nearest the landmark follow up times were utilized. Improvement in VAS after cryoneurolysis was calculated by subtracting post treatment VAS from pretreatment VAS.

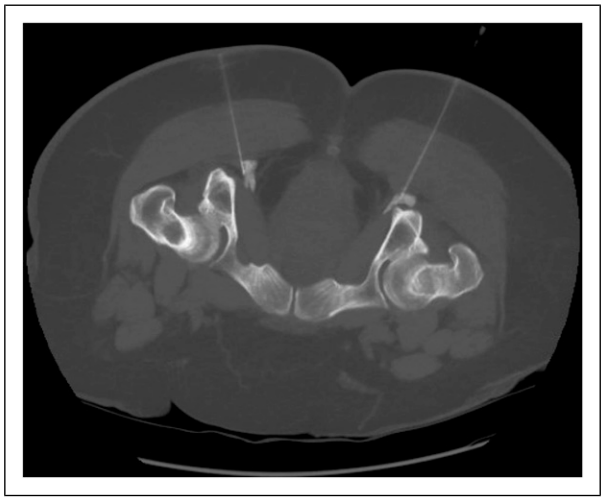


Figure 1. CT-guided bilateral pudendal nerve block

Percentage improvement was calculated with the following equation:

Percent improvement = 100 x ((pretreatment VAS-Post treatment VAS))/(Pretreatment VAS)

AEs were graded according the Common Terminology Criteria for Adverse Events version 6.[10] Given the fundamental difference in patient populations, patients with malignant pain were compared to those without malignant pain; however, 6 month analysis was not performed in this cohort due to sample size. Patient demographics and underlying pelvic pain etiologies are shown in Table 1.

Treatment Protocol

Three fellowship-trained interventional radiologists performed the pudendal nerve cryoneurolysis. Patients were

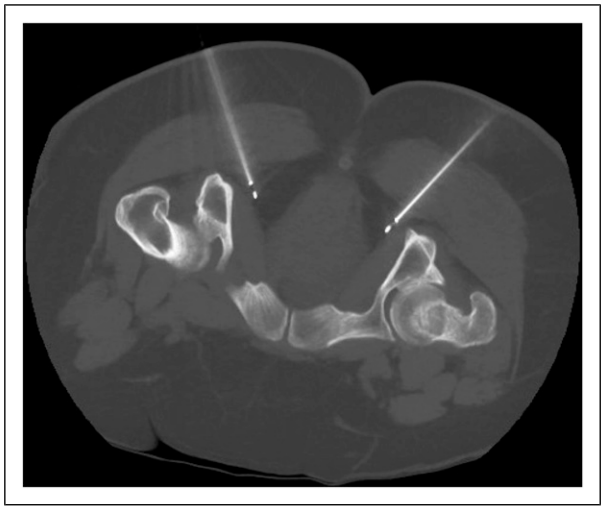


Figure 2. CT-guided bilateral pudendal nerve cryoneurolysis

Table 2. Pain Measurements by Time Point

Variable	Pretreatment	1 week post	1 month post	3 months post	6 months post
VAS	8.9 ± 1.2	3.9 ± 2.1	4.6 ± 2.3	3.9 ± 2.2	6.2 ± 3.4
Morphine equivalents*	50 (range:0-252)	30 (range:0-216)	17.5 (range:0-216)	40 (range:0-216)	85 (range:0-216)
ECOG*	2 (range:1-3)	2 (range:0-3)	2 (range:1-3)	1 (range:1-3)	2 (range:1-4)

VAS = Visual analog scale, ECOG = Eastern Cooperative Oncology Group performance status.

evaluated in clinic or at bedside depending on if they were outpatients or inpatients. All procedures were performed under CT-guidance. All patients were offered a diagnostic/therapeutic block prior to the cryoneurolysis; however, patients who had severe pain (either currently admitted to the hospital or felt to be at risk for admission) and had an obvious cause of pain (ie, pelvic malignancy, etc) were counseled that proceeding straight to cryoneurolysis may be advantageous. CT-guided diagnostic/therapeutic pudendal nerve blocks were performed with a combination of Lidocaine 1%, Bupivacaine 0.25%, and Triamcinolone, and attempted to assess pain response and potential adverse side effects (Figure 1). A pain reduction following the block of ≥3 on the VAS was considered to be significant.[6] Cryoneurolysis of the pudendal nerve was performed using 1 or 2 cryoablation probes (Varian Medical Systems, Palo Alto, CA, USA) positioned within the pudendal canal (Figure 2). Two freeze-thaw cycles were performed consisting of an 8-10 minute freeze cycle followed by a 4 minute passive thaw cycle. The majority of the cryoneurolysis procedures were performed with moderate sedation. General anesthesia was used in a small number of cases.

Statistical Analysis

Continuous variables are reported as mean ± standard deviation or median (Interquartile range (IQR)) and evaluated with paired-t or Mann-Whitney U tests as appropriate. Categorical variables were reported as fractions (percentages) and analyzed with chi-squared or Fisher exact tests as appropriate.

Results

For results concerning pain scores and ME equivalents please see Table 2. When pre-treatment (8.9 ± 1.2) VAS scores were compared to 1 week (3.9 ± 2.1, $P < 0.001$), 1 month (4.6 ± 2.3, $P < 0.001$), 3 months (3.9 ± 2.2, $P < 0.001$) and 6 months post (6.2 ± 3.4, $P < 0.001$) there was significant improvement at all time points. When looking at ME, there was a significant difference when comparing pre-treatment ME (50 (range:0-252)) to 1 week post-treatment (30 (range:0-216), $P = 0.04$) and 1 month post-treatment (17.5 (range:0-216), $P = 0.032$); however, there was no significant difference in ME at 3 months post-treatment (40 (range:0-216), $P = 0.239$) or 6 months post-treatment (85 (range:0-216), $P = 0.452$).

The mean VAS had decreased by 4.4 ± 2.3, 4.0 ± 2.5, 3.8 ± 2.8, and 1.9 ± 3.1 at 1 week, 1 month, 3 months and 6 months post treatment, respectively (Figure 3). Similarly, the percent VAS improvement at 1 week, 1 month, 3 months and 6 months post treatment was 49.2% ± 26.7%, 45.4% ± 27.5%, 42.2% ± 28.1% and 13.5% ± 27.5%, respectively (Figure 4).

When comparing patients with and without malignant pain. There was significant improvement in the VAS score

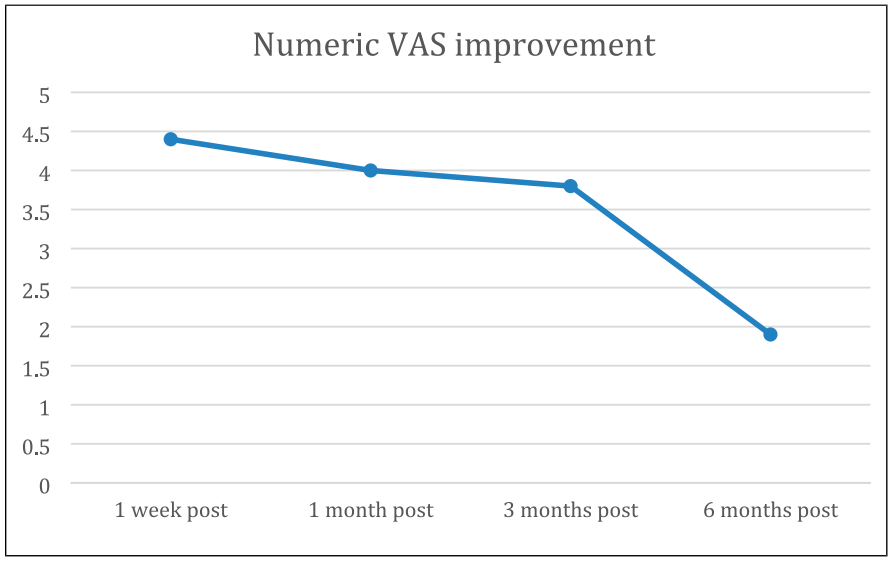


Figure 3. Mean decrease of VAS score a 1 week, 1 month, 3 months, and 6 months post-treatment

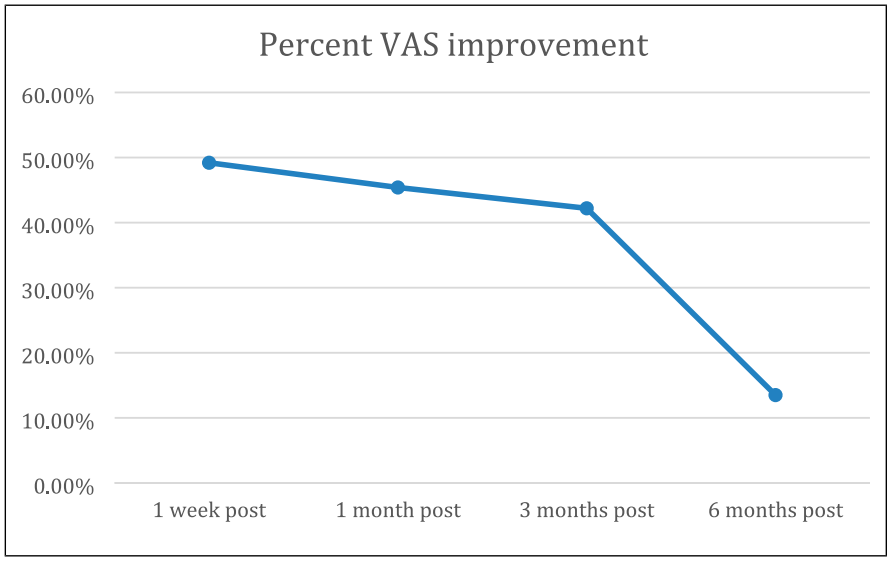


Figure 4. Percent VAS improvement at 1 week, 1 month, 3 months, and 6 months post-treatment

when comparing pre (8.8 ± 1.2) to 1 week (3.8 ± 2.2, $P < 0.001$), 1 month (4.7 ± 2.4, $P < 0.001$) and 3 months (3.8 ± 2.4, $P < 0.001$) post cryoneurolysis in patients with malignant pain. Similarly, in the non-malignant cohort there was significant improvement in VAS scores when comparing pre-treatment (9.5 ± 1) to 1 week (4.5 ± 1.9, $P = 0.008$), 1 month (4.3 ± 1.7, $P = 0.006$) and 3 months (4 ± 1.7, $P = 0.012$) post treatment scores. For further details please see Table 3; however, when comparing numeric improvement and percentage improvement in VAS no significant difference was identified between the malignant and non-malignant cohorts.

No adverse events were reported.

Discussion

This study assesses the safety and efficacy of CT-guided pudendal nerve cryoneurolysis in 30 patients with refractory pelvic pain due to malignant (n = 25) or non-malignant (n = 5) etiologies. The procedure appeared safe, as all patients underwent successful treatment with no reported adverse events. The study demonstrated efficacy through significant improvements in numeric pain scores, percentage pain reduction, and post-procedural morphine equivalents. When comparing improvements in pain scores and percentage reduction between the malignant and non-malignant cohorts, no significant differences were observed, suggesting that

Table 3. Comparison of Malignant and Non-malignant Pain Patients

Variable	Malignant pain	Non-malignant pain	*P* value
Age (years)	59.8 ± 13.1	58.2 ± 11.5	0.399
Sex			1
Male	9 (9/25, 36%)	2 (2/5, 40%)	
Female	16 (16/25, 64%))	3 (3/5, 60%)	
INR	1.2 ± 0.2	1.3 ± 0.2	0.226
Hgb	10.7 ± 3.4	9.5 ± 1.6	0.363
CR	0.8 ± 0.4	1.6 ± 1.3	0.029
Platelets	283.1 ± 146.9	343.3 ± 106.9	0.390
Bilirubin	0.3 (range: 0.2-2)	0.2 (range:0.2-0.3)	0.211
Pretreatment VAS	8.8 ± 1.2	9.5 ± 1	0.270
1 w post num VAS	4.7 ± 2.2	3.8 ± 3.2	0.230
1 w post % improvement	47.8% ± 26.7%	47.5% ± 18.9%	0.185
1 m post num VAS	4.5 ± 2.6	4 ± 3.2	0.349
1 m post % improvement	48% ± 28.1%	41.5% ± 33.7%	0.328
3 m post num VAS	4.8 ± 2.9	3.8 ± 3.8	0.251
3 m post % improvement	47.1% ± 28.4%	38.5% ± 43.8%	0.323

pudendal nerve cryoneurolysis may be effective for both etiologies. However, caution is needed in the interpretation of this data given the small sample size.

To the authors' knowledge, very little data is available comparing the results of present study. Previously, Prologo et al[7] reported CT-guided pudendal cryoneurolysis in 10 patients with refractory pelvic pain due to pelvic neoplasms. The study included only short-term follow-up (12.8 ± 3.7 days) for 5 patients, with a mean follow-up VAS score of 4.0 and a mean pre- to post-treatment difference of 5.2 points. When comparing the short-term results of present study (1 week), with a mean VAS of 3.9 and a mean pre- to post-treatment difference of 5.0 points, the findings are consistent with those of Prologo et al[7] Another study of 11 patients with non-malignant pelvic pain managed with CT-guided cryoneurolysis reported a mean pretreatment VAS score of 7.6, with post-treatment improvements to 2.6 at 24 hours, 3.5 at 45 days, and 3.1 at 6 months.[9] The findings of the present study are consistent with reported outcomes, as significant pain improvement was observed at 1 week, 1 month, 3 months, and 6 months. Additionally, the present study is in line with the available literature, as cryoneurolysis appears to provide effective pain relief for both malignant and non-malignant pelvic pain etiologies.[7,9] No statistically significant differences were found when comparing outcomes between the non-malignant and malignant cohorts.

Cryoneurolysis produces reversible and more controllable axonal injury, with visualization of the ablation zone through real-time ice-ball formation, providing long-term analgesia without permanent nerve destruction.[5,6] This may be advantageous compared with chemical cryoneurolysis and radiofrequency ablation, which typically produce permanent nerve injury and associated with non-target nerve injury, intraprocedural and postprocedural pain, and neuroma formation.[5,11,12] In the present study, significant pain reduction was observed at all time points, including the 6-month follow-up, supporting the durability of this reversible axonal injury. Although morphine equivalents were significantly reduced only at the 1-week and 1-month follow-up points, mean VAS scores and percentage pain reductions remained significantly improved at 3- and 6-month follow-ups likely reflecting partial nerve regeneration but overall sustained pain relief. Nerve regeneration following cryoneurolysis, occurring through Wallerian degeneration, takes approximately 3-6 months for complete nerve regeneration with associated possible return of pre-procedural symptoms.5 Increased morphine equivalents at the 6-month follow-up compared to the pre-treatment morphine equivalents likely represents expected nerve regeneration with associated return of pre-procedural pain requiring increased opioid analgesia for pain mitigation.

Chlorogiannis et al[13] reported significant pain improvement up to 12-month follow-up after cryoneurolysis of the splanchnic nerves for abdominal pain, supporting the meaningful and durable analgesia with cryoneurolysis. Given that most patients (n = 25) in the present cohort underwent the procedure for palliative pain control related to pelvic malignancies, a 6-month duration of pain relief is a clinically meaningful and acceptable therapeutic interval for cancer-associated pelvic pain.

Regarding safety, the present study demonstrated no procedure-related complications and achieved a 100% technical success rate in obtaining precise CT-guided access to the pudendal neurolysis. These findings underscore the value of advanced cross-sectional imaging in pudendal nerve cryoneurolysis, particularly in minimizing non-target ablation and reducing the risk of adjacent neural or soft-tissue injury. Prior case reports have described fluoroscopy-guided techniques that relied on nerve stimulation to confirm pudendal nerve access[14]; however, in patients with pelvic malignancy, tumor-related mass effect or distortion of normal anatomy may limit the accuracy and reliability of these approaches. CT guidance therefore offers a significant advantage by enabling accurate needle trajectory planning, real-time visualization of surrounding structures, and direct monitoring of the ablation zone. Furthermore, a high technical success rate without complications was also reported in previous CT-guided pudendal nerve block and cryoneurolysis studies, further supporting the safety and precision of CT-based pudendal nerve intervention.[7,9,15]

This paper has several limitations. The small sample size and retrospective design are the primary limitations of the present study. Additionally, subgroup sizes were not matched, and the non-malignant cohort was relatively small. Outcomes were self-adjudicated by the study authors, and the VAS score—while widely used—is an imperfect measure of pain because it relies on subjective patient reporting. To partially mitigate this limitation, MEs were also collected to provide a more objective assessment of analgesic requirements. Despite these limitations, to the knowledge of the authors, this study represents the largest cohort to date evaluating CT-guided pudendal nerve cryoneurolysis.

In conclusion, CT-guided pudendal cryoneurolysis appears to be a safe and effective technique capable of providing meaningful pain relief for up to 6 months across both malignant and non-malignant pelvic pain etiologies. Further investigation with larger cohorts is needed to validate the findings of the present study and determine optimum technical parameters for durable and effective pain relief.

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